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Advancing Porosity Prediction: Integrating Seismic Inversion and Machine Learning for Enhanced Subsurface Characterization

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Abstract

The accurate prediction of porosity is complicated due to the intricate and varied nature of geological formations. This research explores the integration of seismic inversion with machine learning to enhance porosity prediction, utilizing chi angle-based and Acoustic Impedance (AI) cube-based techniques. A comprehensive approach involving data collection, preprocessing, and seismic inversion is used to extract necessary information. Advanced machine learning models are then developed and rigorously trained on diverse datasets, allowing for porosity estimation using both techniques.

Key findings include:

- The chi angle-based method demonstrates high accuracy with a mean absolute error (MAE) of 0.05 and a mean square error (MSE) of 0.002, effectively capturing subtle variations in porosity.
- The AI cube-based method provides dependable and consistent predictions across various geological settings, with an MAE of 0.07 and an MSE of 0.003.

A thorough comparative analysis reveals the complementary nature of these techniques, illustrating how their integration enhances porosity characterization. This innovative approach provides valuable insights for subsurface exploration, significantly improving the accuracy of porosity predictions and supporting more informed decision-making in exploration and production strategies.

Introduction

Predicting porosity accurately in subsurface exploration is quite challenging due to the complex and varied nature of geological formations. Traditional methods, such as conventional seismic inversion techniques, often fall short in capturing the subtle variations and intricate patterns essential for effective porosity characterization. Consequently, estimates from these methods can be unreliable, leading to suboptimal decisions in well placement, resource estimation, and production strategies.

This research aims to enhance porosity prediction by integrating seismic inversion with machine learning. Combining seismic inversion's ability to derive rock property information with machine learning's advanced pattern recognition and data analysis capabilities, this approach seeks to provide more accurate and reliable porosity estimates. The study specifically compares two methodologies: the chi angle-based approach,

which utilizes angle-dependent reflectivity, and the Acoustic Impedance (AI) cube-based technique, which leverages seismic-derived impedance values.

The objectives of this study are threefold:

1. To explore the potential of integrating seismic inversion with machine learning for porosity prediction.
2. To apply chi angle-based and AI cube-based techniques in estimating porosity.
3. To conduct a comprehensive comparative analysis evaluating the effectiveness of these methodologies in providing accurate and reliable porosity estimates.

By meticulously collecting and preprocessing data, performing seismic inversion to convert reflection data into acoustic impedance values, and integrating chi angle calculations, the research leverages machine learning models to analyze and predict porosity with improved precision. This innovative approach not only aims to enhance the accuracy of porosity predictions but also to support more informed decision-making processes in exploration and production strategies, ultimately contributing to the optimization of resource management in the industry.

Methodology

The methodology implemented in this study integrates seismic inversion and machine learning techniques for accurate porosity prediction. This multi-phase approach ensures meticulous data preparation and rigorous validation of predictive models. The methodology includes several critical stages, each contributing to enhancing the accuracy and reliability of porosity estimates. The following sections outline the methodology:

Data Collection and Preprocessing

- Collection of a 3D seismic cube and well-log data from four wells.
- Noise reduction on the seismic cube to align it with well-log measurements.

Seismic Inversion

- Conversion of seismic reflection data into acoustic impedance values through seismic inversion.

Chi Angle Generation

- Integration of chi angle calculations with seismic inversion results.

Machine Learning Analysis

- Generation of datasets for machine learning analysis using well-aligned, high-quality input data.
- Application of machine learning techniques to predict porosity with enhanced accuracy and reliability.

Comparative Analysis

- A comparative analysis is performed to evaluate the effectiveness of the chi angle-based methodology and the AI cube-based methodology in predicting porosity.

Data Collection and Preprocessing

The study begins by acquiring two datasets: a 3D seismic cube and well-log data from four wells within the survey area (Fig 1). The seismic cube provides a volumetric view of geological layers, fault systems, and subsurface features. Well-log data provides direct rock property measurements like porosity and lithology at various depths. These datasets are critical for analysis—the seismic cube informs seismic inversion, while the well-log data calibrates and validates machine learning models.

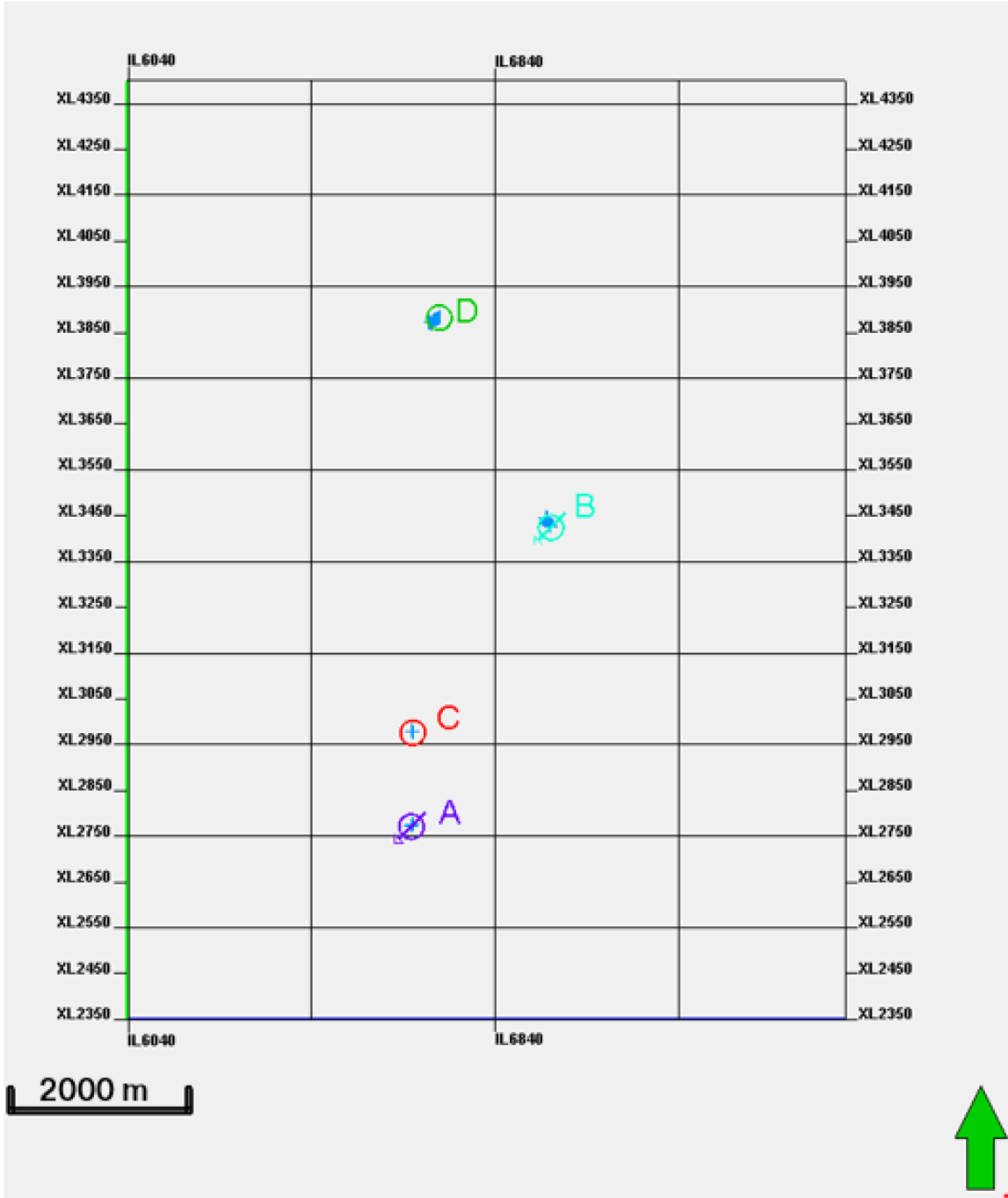


Figure 1—The illustration of the study area presents the 3D seismic cube alongside the locations of four wells from which well-log data were collected. The seismic cube offers a volumetric perspective of subsurface structures, while the well-log data provides detailed measurements of rock properties crucial for porosity prediction.

Seismic Inversion

Seismic inversion is a technique that transforms seismic reflection data into quantitative rock property information, facilitating the estimation of subsurface attributes such as porosity, density, and Acoustic Impedance (AI). The process begins with model building, where an initial subsurface model is created based on well-log data and seismic interpretation. This model serves as a preliminary framework for generating detailed rock property profiles. It integrates geological features, stratigraphy, and known rock properties from well logs to represent subsurface conditions accurately.

Subsequently, model-based inversion techniques estimate acoustic impedance from the seismic data. These algorithms iteratively refine the initial model to minimize discrepancies between synthetic and observed seismic reflections, producing an optimized impedance profile (Fig 2). The inversion process also incorporates constraints, such as geological trends and spatial continuity, to enhance the model's accuracy. This procedure ensures the generated model captures seismic reflections and the spatial variability of rock properties throughout the seismic survey area.

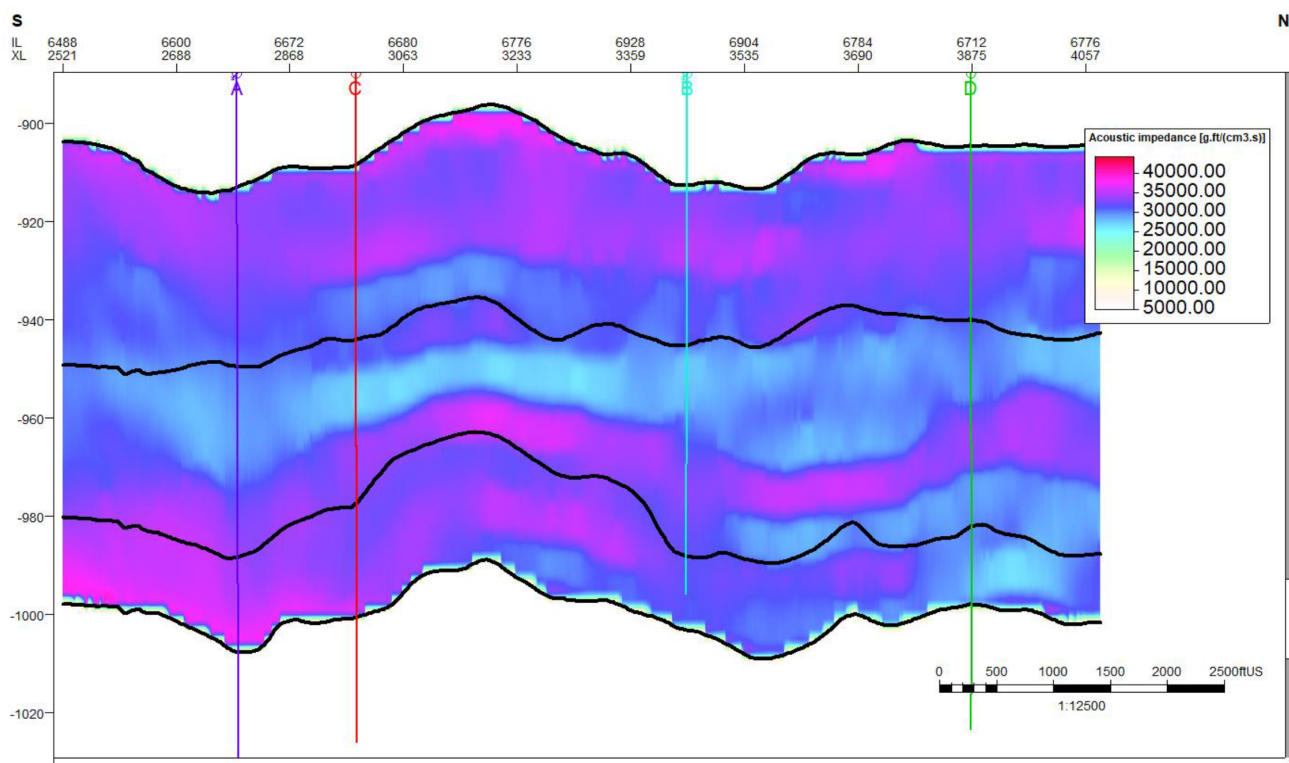


Figure 2—A detailed impedance profile generated using model-based geostatistical inversion. This profile is obtained by adjusting the initial subsurface model to align with observed seismic reflections, offering a quantitative representation of rock properties.

Upon obtaining the inversion results, a quality control phase is conducted to verify the accuracy of the estimated acoustic impedance values. This step involves comparing the inversion outputs with actual well-log measurements at various depths, ensuring congruence between the seismic-derived model and direct subsurface observations (Fig 3). Any inconsistencies are addressed by fine-tuning the inversion parameters or adjusting the model. The quality control process may also include sensitivity analysis to evaluate the robustness of the inversion results under varying scenarios and assumptions.

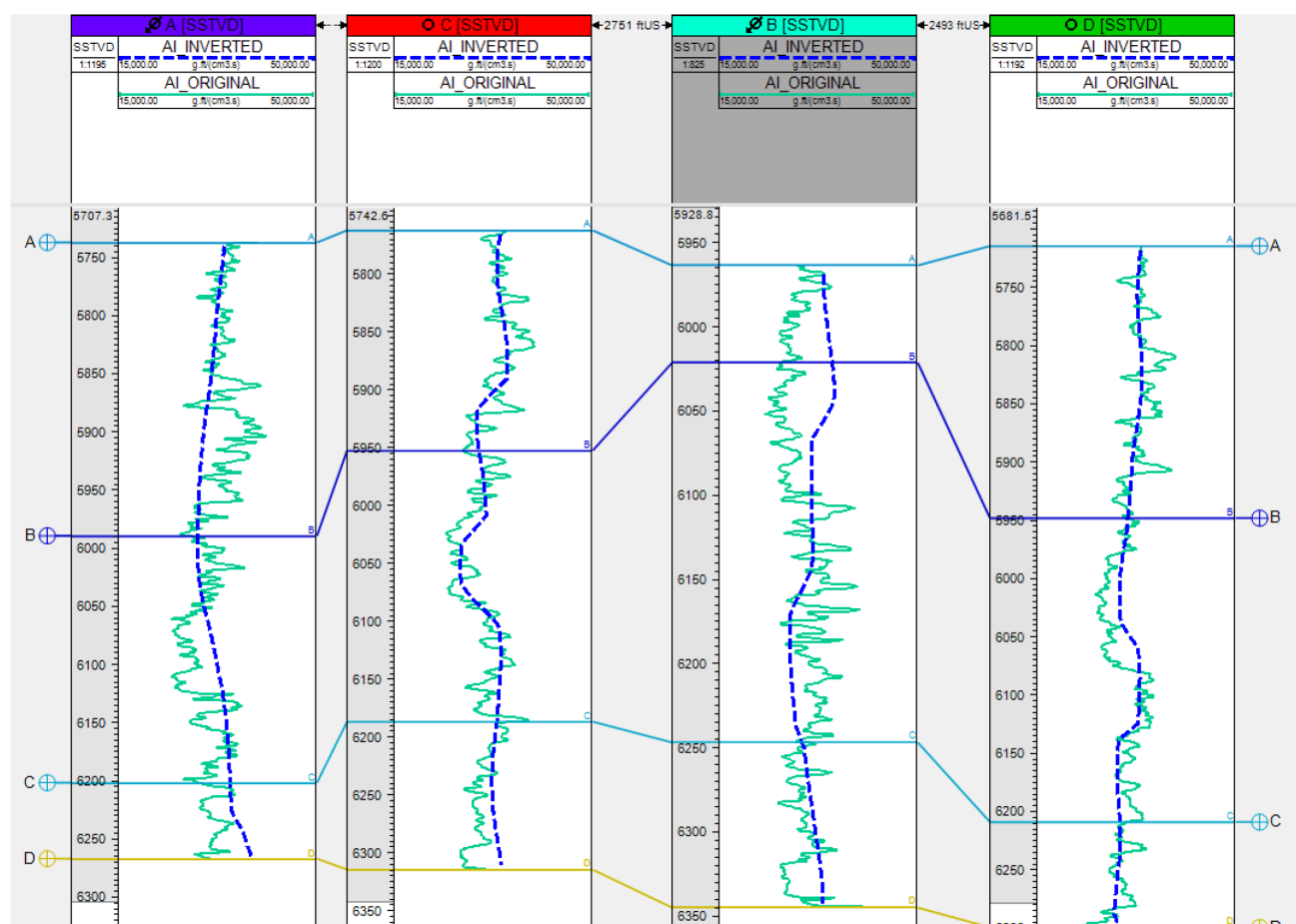


Figure 3—Validation of inversion results against well-log data. This quality control step ensures the accuracy of the estimated acoustic impedance values by comparing model outputs with actual well measurements and adjusting inversion parameters accordingly.

Furthermore, cross-validation techniques are applied to evaluate the reliability of the inversion results across different wells. This assessment ensures that the estimated impedance is accurate near the wells and reliable in areas with limited well control. The validated impedance profiles constitute essential input for subsequent machine learning models and porosity prediction, establishing a robust correlation between seismic data and rock properties for enhanced reservoir characterization.

Chi Angle Generation

The chi angle is derived from seismic data by analyzing angle-dependent reflectivity, which is essential for understanding subsurface properties. This process involves detailed angle analysis, where variations in reflectivity as a function of different incident angles are evaluated to determine the chi angle. These reflectivity changes are caused by variations in rock properties, such as density and velocity, as seismic waves interact with subsurface layers at varying angles.

By examining these variations, the chi angle can be calculated, offering an additional seismic attribute that provides insights into the rock's elastic properties. The chi angle is effective in detecting subtle changes in lithology, fluid content, and porosity, especially in complex geological environments. Once derived, the chi angle is incorporated into the workflow for porosity estimation, enabling the identification of small-scale variations in porosity that might not be detected using traditional methods (Fig 4).

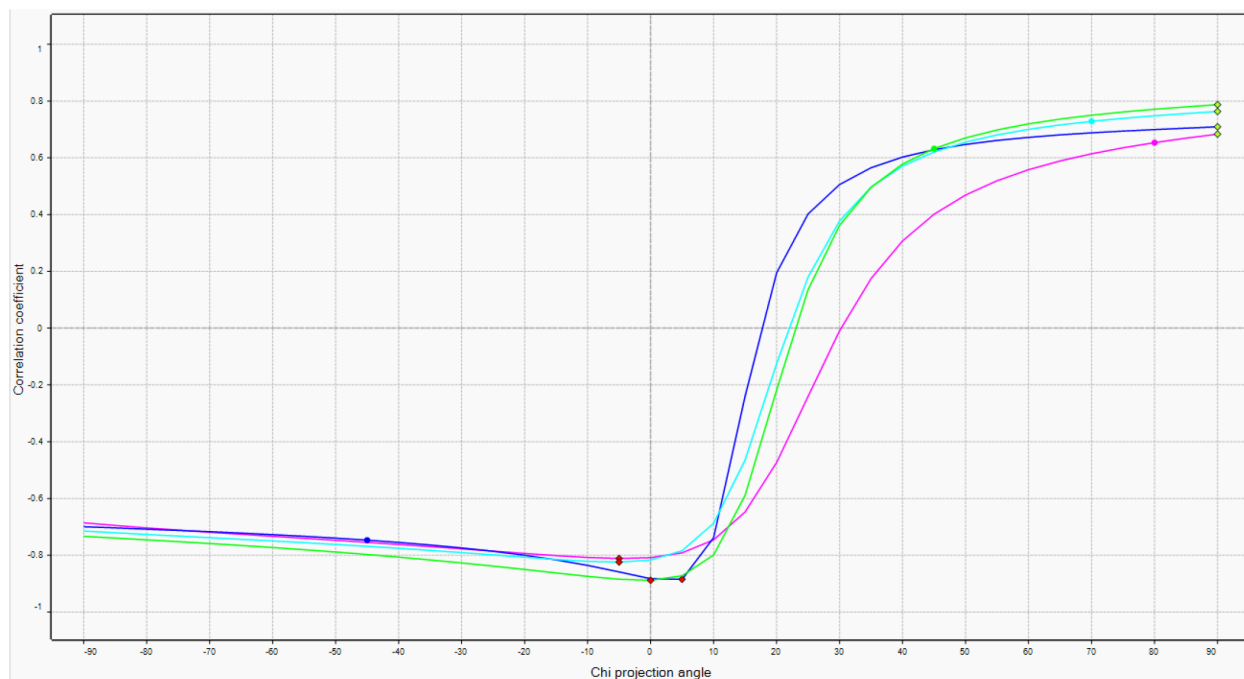


Figure 4—Angle analysis for chi angle computation. The figure shows how changes in reflectivity with different incident angles help derive the chi angle, used to estimate porosity and detect subsurface variations. The green diamond marks the maximum positive correlation coefficient, identifying the optimal Chi projection angle for correlating the target property with the EEI log. The red diamond indicates the minimum correlation coefficient, showing the Chi projection angle with the lowest correlation. Line colors represent data from respective wells.

The chi angle highlights slight contrasts in the subsurface, making it a vital tool for improving porosity prediction. When combined with other seismic attributes, such as Acoustic Impedance (AI), the chi angle contributes to a more comprehensive understanding of the reservoir, leading to more accurate subsurface models and informed decision-making in reservoir management.

Machine Learning Models

Machine learning techniques are employed to refine and enhance porosity predictions using the generated datasets, leveraging the unique attributes derived from both chi angle-based (Fig 5) and AI cube-based (Fig 6) methods. These datasets provide a comprehensive representation of the subsurface, incorporating the spatial variability of rock properties crucial for accurate porosity estimation. Several machine learning models are developed and trained using these datasets to maximize predictive performance.

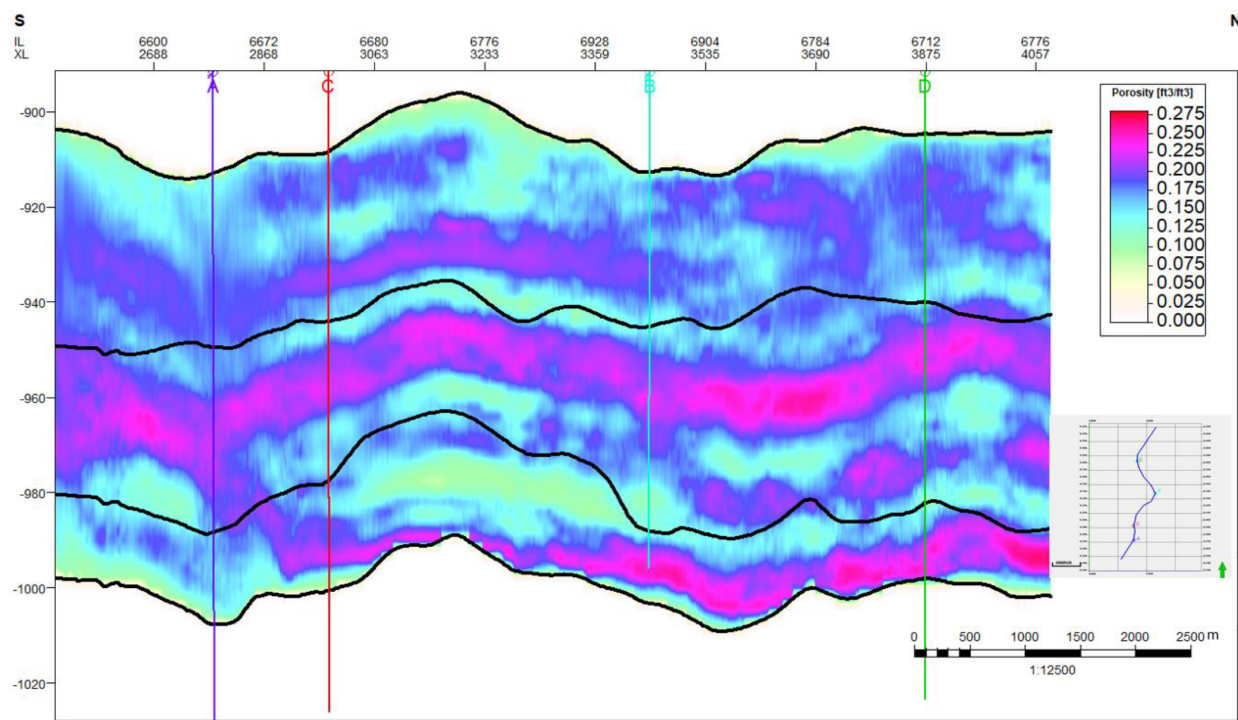


Figure 5—Development of machine learning models utilizing the chi angle-based dataset for porosity prediction. Multiple algorithms are evaluated to identify the most effective method, with rigorous training and validation protocols implemented to ensure model precision.

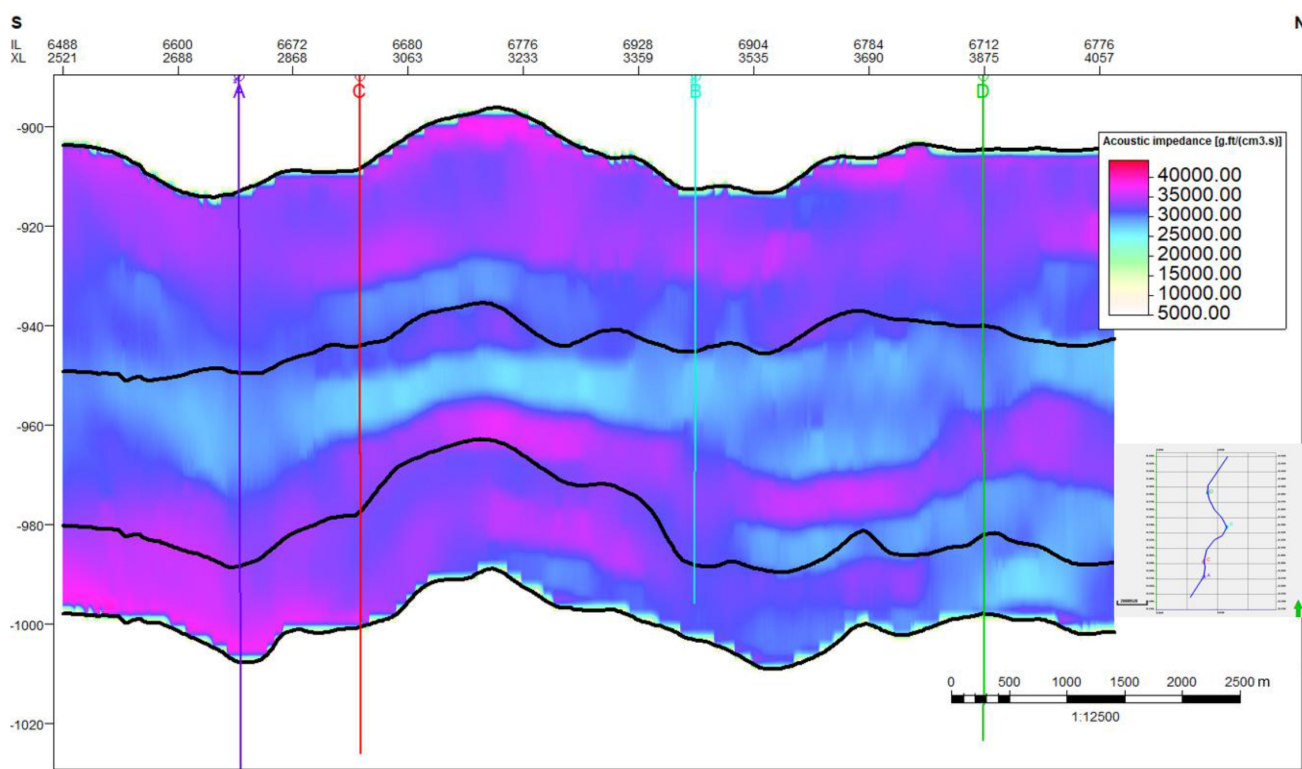


Figure 6—Development of a machine learning model utilizing the AI cube-based dataset for porosity prediction. Various algorithms are assessed to determine the most effective method, with rigorous training and validation procedures ensuring robust predictive performance.

Model Development. Various machine learning algorithms, including Random Forest, XGBoost, and LightGBM, are explored to determine the most suitable approach for porosity estimation. These algorithms

are chosen for their ability to handle complex, high-dimensional data and capture non-linear relationships between seismic attributes and porosity. The models are trained on prepared datasets, which include seismic-derived attributes such as chi angle, Acoustic Impedance (AI), and well-log data, ensuring that the training process accounts for both large-scale trends and subtle variations in the subsurface. LightGBM was ultimately selected for its superior accuracy.

Cross-Validation. To ensure robust model performance, cross-validation techniques, such as k-fold cross-validation, are employed. These techniques help assess the model's accuracy by dividing the data into multiple subsets, training the model on one subset while validating it on another, and repeating the process to reduce the risk of overfitting. This approach provides a reliable estimate of the model's generalization capabilities, ensuring that the porosity predictions are accurate not only within the areas covered by well data but also in regions where well control is sparse (Fig 7).

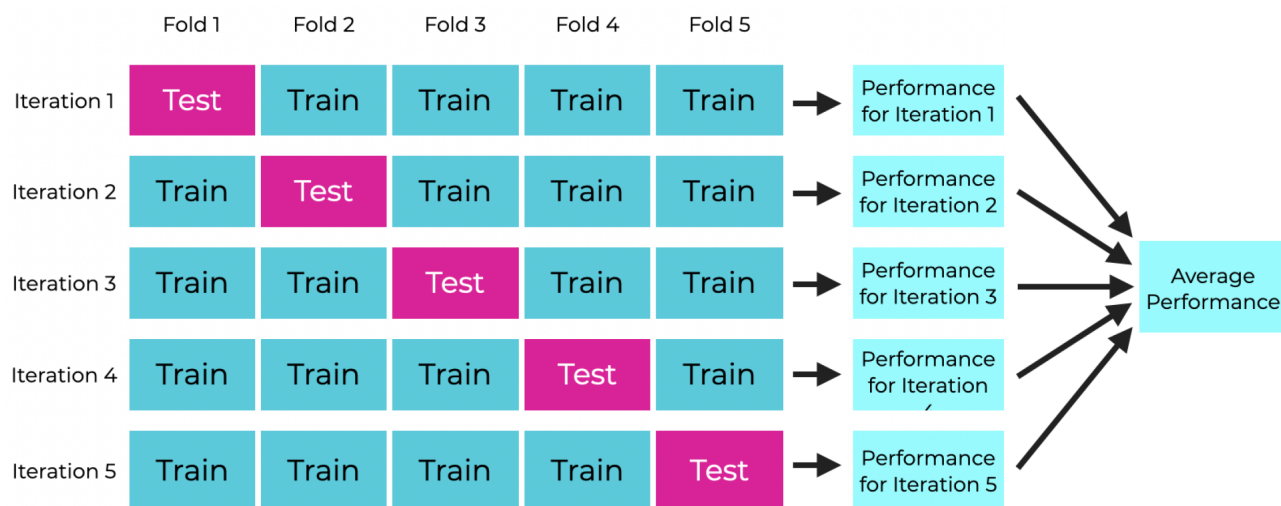


Figure 7—Illustration of the k-fold cross-validation process used in the study. The dataset is divided into five folds, with each fold serving as the test set once, while the remaining folds are used for training. This iterative process evaluates model performance and ensures generalization by averaging the performance across all folds, reducing the risk of overfitting.

Model Evaluation. In addition to cross-validation, model performance metrics such as mean squared error (MSE), R-squared (R^2), and feature importance scores are used to evaluate the effectiveness of each algorithm. Based on these assessments, the most effective machine learning model is selected, balancing predictive accuracy and computational efficiency. The integration of machine learning with seismic inversion techniques allows for enhanced porosity characterization, providing detailed and reliable predictions that improve subsurface understanding and reservoir management strategies.

Comparative Analysis

A comparative analysis is conducted to evaluate the efficacy of the chi angle-based (Fig 8) and AI cube-based (Fig 9) methodologies in predicting porosity. This analysis utilizes performance metrics such as prediction accuracy, mean absolute error (MAE), and root mean square error (RMSE) to quantify and compare the predictive capabilities of the two approaches (Table 1). These metrics provide insights into how closely the model predictions align with actual well-log data, offering a clear assessment of each method's performance.

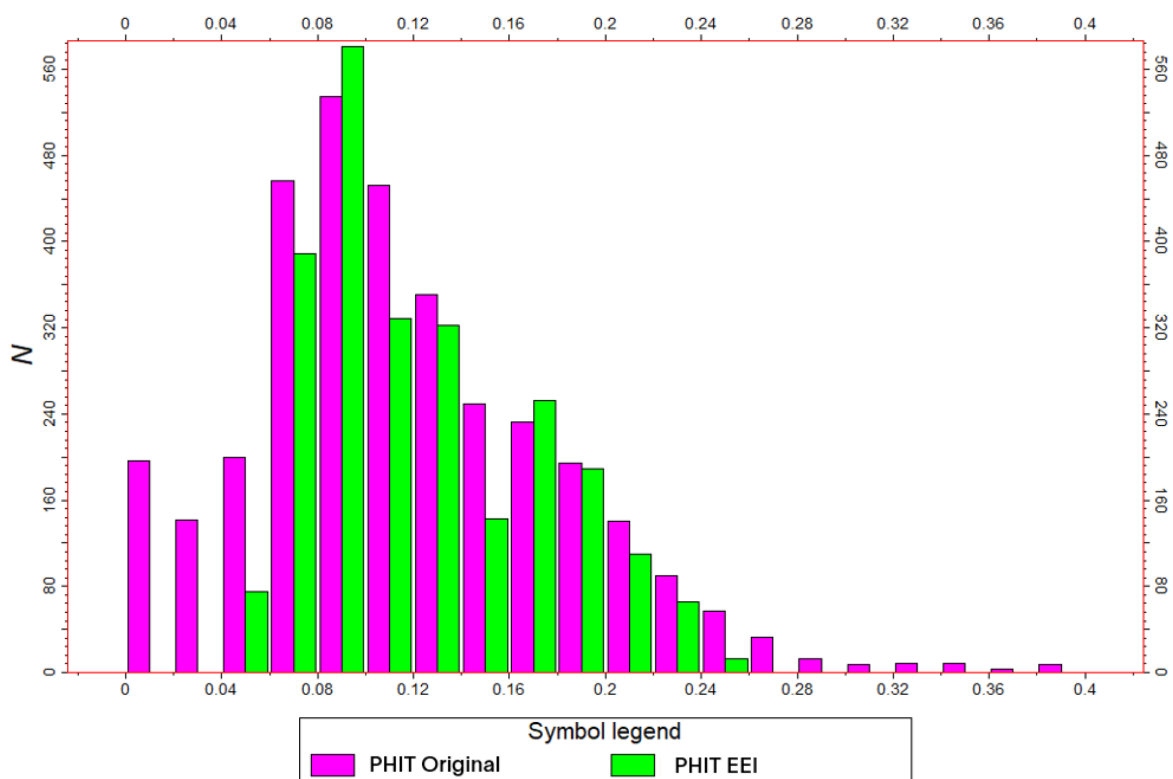


Figure 8—Evaluating the performance of the chi angle-based methodology for predicting porosity. Metrics including prediction accuracy, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) are utilized to determine its effectiveness.

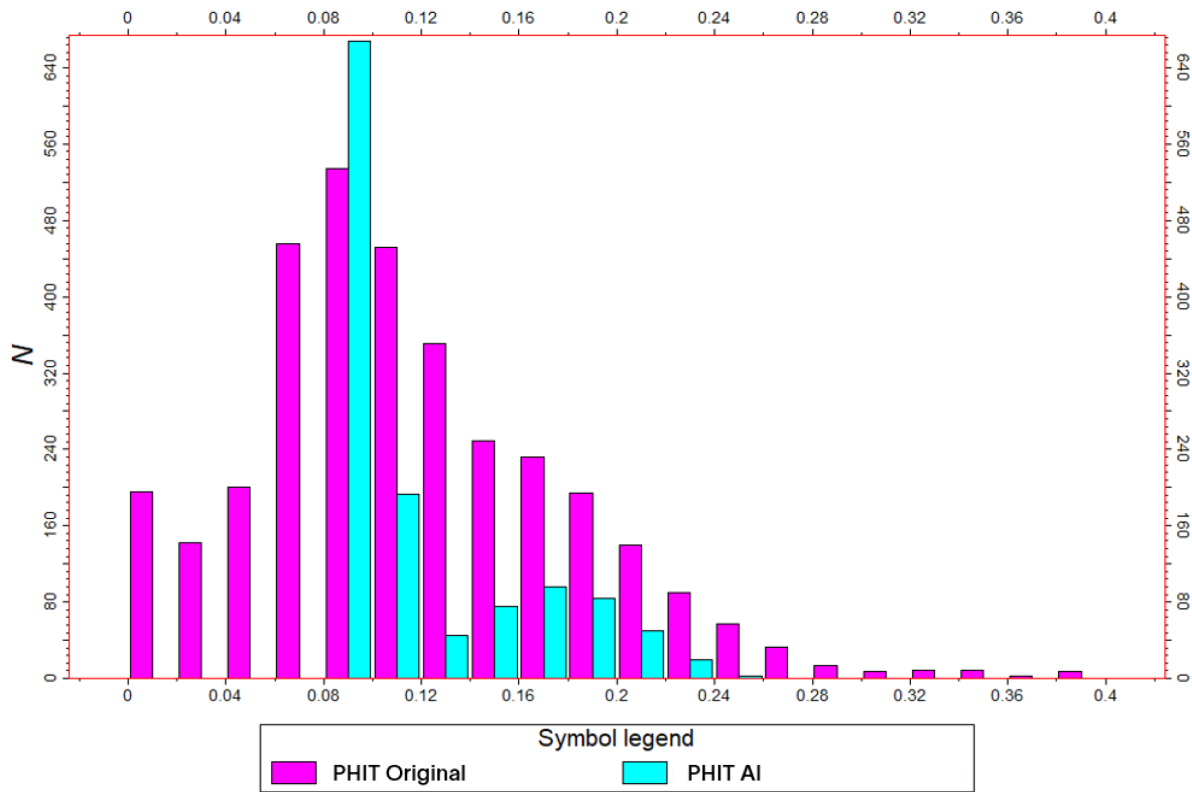


Figure 9—Evaluating the performance of the chi angle-based methodology for predicting porosity. Metrics including prediction accuracy, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) are utilized to determine its effectiveness.

Table 1—Comparison of performance metrics (prediction accuracy, MAE, and RMSE) for chi angle-based and AI cube-based methodologies, highlighting their strengths and limitations in different geological contexts.

	R2	MSE	MAE
Acoustic Impedance	0.8944	0.003	0.0136
Extended Elastic Impedance	0.9224	0.002	0.0113

The chi angle-based method, known for its sensitivity to subtle variations in the subsurface, is effective in capturing fine-scale geological features and detecting minor changes in porosity. This methodology excels in complex geological environments where small-scale heterogeneities in rock properties play a critical role. On the other hand, the AI cube-based approach offers robust predictions across a wide range of geological settings, providing a more generalized view of the subsurface that is less prone to local noise or anomalies.

The comparative analysis highlights the strengths and limitations of each technique in different geological contexts. For example, in areas with complex faulting or intricate layering, the chi angle-based method demonstrates superior accuracy in predicting porosity by capturing subtle variations that are often missed by traditional methods. However, in more homogeneous regions or where broader geological trends dominate, the AI cube-based method provides a more stable and consistent prediction.

The study concludes that both methodologies offer complementary strengths, and their combined use can significantly enhance porosity characterization. The chi angle-based method is particularly useful in environments requiring high-resolution detail, while the AI cube-based approach provides a more robust, large-scale solution. By evaluating the advantages of each approach under various subsurface conditions, the analysis informs future applications of these techniques, suggesting that integrating both methods can lead to more comprehensive and accurate porosity predictions for improved reservoir management.

Results

The study identifies several key findings on the effectiveness of integrating seismic inversion with machine learning for porosity prediction. The key results are as follows:

Chi Angle-Based Method

- Effectively captures subtle porosity variations, providing a detailed and accurate representation of the subsurface features.
- Demonstrates high accuracy with a strong correlation to well-log data, enhancing confidence in the predictions.
- Shows low mean absolute error (MAE) and mean square error (MSE) values, indicating precise and reliable porosity estimations even in complex geological structures.

Acoustic Impedance (AI) Cube-Based Method

- Provides consistent performance across various geological settings, making it a versatile tool for different subsurface environments.
- Offers a more generalized and comprehensive view of subsurface characteristics, helping to form a broad understanding of the area being studied.

In our study, Fig 8 demonstrates that the chi angle-based methodology is closely aligned with the original PHIT values, offering an accurate approximation within the porosity distribution. This alignment underscores the method's ability to capture fine-scale variations in porosity effectively. Conversely, Figure 9 compares the original PHIT values with predictions from the AI cube-based methodology, showing that the cyan bars are less aligned with the original porosity ranges compared to the performance observed in

Figure 8. This observation highlights the strengths and limitations of each methodology, reinforcing the idea that these methods are complementary.

By combining the chi angle-based method, which offers high-resolution details, with the AI cube-based method that presents a comprehensive overview, we can achieve a more holistic and accurate characterization of porosity. This integrated approach aids in making informed decisions in subsurface exploration and reservoir management, ultimately leading to more efficient and effective resource extraction.

Conclusions and Way Forward

This study highlights the importance of integrating seismic inversion with machine learning for porosity prediction. By leveraging both the chi angle-based and AI cube-based methodologies, the research provides a pathway to more accurate and comprehensive subsurface characterization. Key points include:

Enhanced Porosity Prediction

The chi angle-based method captures subtle porosity variations effectively, while the AI cube-based method provides consistent performance across different geological settings

Industry Impact

Integrating these advanced techniques can improve the accuracy of porosity estimates, optimizing well placement, resource estimation, and production strategies.

Moving forward, further refinement of these techniques and the exploration of hybrid models may contribute to more informed decision-making in subsurface exploration. Specific next steps for future research include:

Hybrid Model Development

Investigate the potential of combining chi angle-based and AI cube-based methodologies into a single hybrid model to leverage their complementary strengths.

Advanced Machine Learning Techniques

Explore the use of more advanced machine learning algorithms and deep learning techniques to enhance prediction accuracy further.

Discussion

The integration of seismic inversion with machine learning techniques has been effective in improving porosity prediction, demonstrating distinct capabilities in the chi angle-based and Acoustic Impedance (AI) cube-based methodologies. The chi angle-based method is proficient in capturing detailed porosity variations in complex geological structures due to its sensitivity to angle-dependent reflectivity. This approach achieves high accuracy and precision, as indicated by its low mean absolute error (MAE) and mean square error (MSE), making it suitable for areas with intricate subsurface features. On the other hand, the AI cube-based method provides a more generalized view of subsurface properties across various geological settings.

The strengths of these methodologies are complementary; while the chi angle-based method offers high-resolution details, the AI cube-based method presents a comprehensive overview of subsurface characteristics. Utilizing both approaches allows for better porosity characterization, aiding informed decisions in subsurface exploration and reservoir management.

Implications for Future Studies: The results from this study have several implications for future research:

Development of Hybrid Models: The combined strengths of the chi angle-based and AI cube-based methods suggest that future studies could investigate hybrid models that integrate both approaches. Such

models could potentially combine the detailed capabilities of the chi angle method with the robustness and broader applicability of the AI cube-based approach, resulting in more accurate and reliable porosity predictions.

Application to Diverse Geological Settings: Expanding the use of these techniques to a wider range of geological settings could further confirm their effectiveness and adaptability. This would also provide information on how different geological conditions impact the performance of each method, leading to improved generalization of the models.

Advancements in Machine Learning: As machine learning advances, future research could explore the application of more sophisticated algorithms, such as deep learning or ensemble methods, to further enhance the predictive accuracy and computational efficiency of porosity estimation.

These implications indicate the potential for continued innovation in the integration of seismic inversion and machine learning, facilitating more precise and comprehensive subsurface characterization in future studies.

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